

Fraunhofer IIS, Germany. It has been tested in Danish Superliga and at 2012 Club World Cup. It is a radio based system that uses electromagnetic induction to determine whether or not the whole ball has passed over the goal line.

A passive electronic circuit is embedded into the ball between the leather outer layer and the inflatable centre. An electromagnetic field is created in the goal-mouth and its strength is monitored by a computer. When the circuit in the ball crosses the goal line, the electromagnetic field changes and processed data is transmitted wirelessly to the referee's wristwatch. A message is displayed in real time, in addition to a vibrating alert.

Because the technology behind



Fig. 7: Goal Ref system's electronic circuit installed on the posts of the goal

Goal Ref is inexpensive compared to Hawk-Eye, it is easier to implement at the lower levels of the game, addressing the concerns that goal line technology would exacerbate the gulf between grassroots and megabucks.

Future scope

With the amount of money flowing in football, the pressure to succeed is increasing drastically. Fans are

demanding FIFA to introduce technologies to tackle issues such as wrong offside decisions and diving. Introduction of video technology for decisions regarding penalties, red cards and identity crisis is set to be put on trial mode.

Kit makers are planning to intro-

duce textile electronics to build the detection system in their kits. Along with medical sensing technology, referees could catch those players who waste time rolling around acting as if they are injured. Ball impact sensors and accelerometers in football boots could be introduced in the future, which would measure every step, strength and angle of impact of the ball, or another player's leg. **EFY**



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